

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
#install.packages('dplyr')
#install.packages('lubridate')
#install.packages('here')

library(here)
library(tidyverse)
library(lubridate)

setwd("/home/guest/EDE_Fall2025")

getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Neonics <- read.csv(  
  file = here("./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),  
  stringsAsFactors = TRUE  
)  
  
Litter <- read.csv(  
  file = here("./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),  
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might be interested in the ecotoxicology of neonicotinoids on insects because the insecticides could be harmful to more insects than it was originally intended. Insects could also be developing an immunity to neonicotinoids and we would need to know that.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: We would be interested in the litter of a forest because when they decompose they are becoming the soil and are providing nutrients for the plants and trees that reside there. To monitor the health of a forest, one thing we need to know is what is contributing to the soil and its nutrients.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. 2 types litter traps are set up, one on the ground and one suspended in the air 2. The plots they observe are 40mx40m 3. They need one pair of suspended and grounded traps every 400 m² so that usually means 1-4 pair traps per plot

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
summary(Neonics)
```

```

## CAS.Number
## Min. : 58842209
## 1st Qu.:138261413
## Median :138261413
## Mean :147651982
## 3rd Qu.:153719234
## Max. :210880925
##
##
## Chemical.Name
## (2E)-1-[(6-Chloro-3-pyridinyl)methyl]-N-nitro-2-imidazolidinimine :2658
## 3-[(2-Chloro-5-thiazolyl)methyl]tetrahydro-5-methyl-N-nitro-4H-1,3,5-oxadiazin-4-imine: 686
## [C(E)]-N-[(2-Chloro-5-thiazolyl)methyl]-N'-methyl-N''-nitroguanidine : 452
## (1E)-N-[(6-Chloro-3-pyridinyl)methyl]-N'-cyano-N-methylethanimidamide : 420
## N''-Methyl-N-nitro-N'-[(tetrahydro-3-furanyl)methyl]guanidine : 218
## [N(Z)]-N-[3-[(6-Chloro-3-pyridinyl)methyl]-2-thiazolidinylidene]cyanamide : 128
## (Other) : 61
##
## Chemical.Grade
## Not reported :3989
## Technical grade, technical product, technical formulation: 422
## Pestanal grade : 93
## Not coded : 53
## Commercial grade : 27
## Analytical grade : 15
## (Other) : 24
##
## Chemical.Analysis.Method
## Measured : 230
## Not coded : 51
## Not reported : 5
## Unmeasured :4321
## Unmeasured values (some measured values reported in article): 16
##
##
## Chemical.Purity Species.Scientific.Name
## NR :2502 Apis mellifera : 667
## 25 : 244 Bombus terrestris : 183
## 50 : 200 Apis mellifera ssp. carnica : 152
## 20 : 189 Bombus impatiens : 140
## 70 : 112 Apis mellifera ssp. ligustica: 113
## 75 : 89 Popillia japonica : 94
## (Other):1287 (Other) :3274
##
## Species.Common.Name
## Honey Bee : 667
## Parasitic Wasp : 285
## Buff Tailed Bumblebee: 183
## Carniolan Honey Bee : 152
## Bumble Bee : 140
## Italian Honeybee : 113
## (Other) :3083
##
## Species.Group
## Insects/Spiders :3569
## Insects/Spiders; Standard Test Species : 27
## Insects/Spiders; Standard Test Species; U.S. Invasive Species: 667
## Insects/Spiders; U.S. Invasive Species : 360
##

```



```
## Williamson,S.M., S.J. Willis, and G.A. Wright : 93
## Laurino,D., A. Manino, A. Patetta, and M. Porporato : 88
## Scholer,J., and V. Krischik : 82
## (Other) :3956
## Reference.Number
## Min. : 344
## 1st Qu.:108459
## Median :165559
## Mean :142189
## 3rd Qu.:168998
## Max. :180410
##
##
## Long-Term Effects of Imidacloprid on the Abundance of Surface- and Soil-Active Nontarget Fauna in T
## Reduced Risk Insecticides to Control Scale Insects and Protect Natural Enemies in the Production and
## Effects of Sublethal Doses of Acetamiprid and Thiamethoxam on the Behavior of the Honeybee (Apis me
## Exposure to Neonicotinoids Influences the Motor Function of Adult Worker Honeybees
## Toxicity of Neonicotinoid Insecticides on Different Honey Bee Genotypes
## Chronic Exposure of Imidacloprid and Clothianidin Reduce Queen Survival, Foraging, and Nectar Storing
## (Other)
##
## Source Publication.Year
## Agric. For. Entomol.11(4): 405-419 : 200 Min. :1982
## Environ. Entomol.41(2): 377-386 : 100 1st Qu.:2005
## Arch. Environ. Contam. Toxicol.54(4): 653-661: 96 Median :2010
## Ecotoxicology23:1409-1418 : 93 Mean :2008
## Bull. Insectol.66(1): 119-126 : 88 3rd Qu.:2013
## PLoS One9(3): 14 p. : 82 Max. :2019
## (Other) :3964
## Summary.of.Additional.Parameters
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active ingred
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active ingred
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active ingred
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active ingred
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Active ingred
## Purity: \xca NR - NR | Organism Age: \xca NR - NR Not reported | Conc 1 (Author): \xca Formulation I
## (Other)
```

#168x30

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect)
```

```
## Accumulation Avoidance Behavior Biochemistry
## 12 102 360 11
## Cell(s) Development Enzyme(s) Feeding behavior
## 9 136 62 255
## Genetics Growth Histology Hormone(s)
## 82 38 5 1
## Immunological Intoxication Morphology Mortality
## 16 12 22 1493
```

##	Physiology	Population	Reproduction
##	7	1803	197

Answer: Mortality, population, behavior, and feeding behavior are the most commonly studied. These are of interest because we need to observe the more dangerous effects of the neonicotinoids. We need to see how it effects mortality in the individual, and how it effects the population as a whole.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name)
```

##	Honey Bee	Parasitic Wasp
##	667	285
##	Buff Tailed Bumblebee	Carniolan Honey Bee
##	183	152
##	Bumble Bee	Italian Honeybee
##	140	113
##	Japanese Beetle	Asian Lady Beetle
##	94	76
##	Euonymus Scale	Wireworm
##	75	69
##	European Dark Bee	Minute Pirate Bug
##	66	62
##	Asian Citrus Psyllid	Parastic Wasp
##	60	58
##	Colorado Potato Beetle	Parasitoid Wasp
##	57	51
##	Erythrina Gall Wasp	Beetle Order
##	49	47
##	Snout Beetle Family, Weevil	Sevenspotted Lady Beetle
##	47	46
##	True Bug Order	Buff-tailed Bumblebee
##	45	39
##	Aphid Family	Cabbage Looper
##	38	38
##	Sweetpotato Whitefly	Braconid Wasp
##	37	33
##	Cotton Aphid	Predatory Mite
##	33	33
##	Ladybird Beetle Family	Parasitoid
##	30	30
##	Scarab Beetle	Spring Tiphia
##	29	29
##	Thrip Order	Ground Beetle Family
##	29	27
##	Rove Beetle Family	Tobacco Aphid
##	27	27
##	Chalcid Wasp	Convergent Lady Beetle
##	25	25

##	Stingless Bee	Spider/Mite Class
##	25	24
##	Tobacco Flea Beetle	Citrus Leafminer
##	24	23
##	Ladybird Beetle	Mason Bee
##	23	22
##	Mosquito	Argentine Ant
##	22	21
##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14
##	Ox Beetle	Red Scale Parasite
##	14	14
##	Spined Soldier Bug	Armoured Scale Family
##	14	13
##	Diamondback Moth	Eulophid Wasp
##	13	13
##	Monarch Butterfly	Predatory Bug
##	13	13
##	Yellow Fever Mosquito	Braconid Parasitoid
##	13	12
##	Common Thrip	Eastern Subterranean Termite
##	12	12
##	Jassid	Mite Order
##	12	12
##	Pea Aphid	Pond Wolf Spider
##	12	12

##	Spotless Ladybird Beetle	Glasshouse Potato Wasp
##	11	10
##	Lacewing	Southern House Mosquito
##	10	10
##	Two Spotted Lady Beetle	Ant Family
##	10	9
##	Apple Maggot	(Other)
##	9	670

Answer: The six most commonly studied speices are honey bee, parasitic wasp, buff tailed bumblebee, carniolan honey bee, bumble bee, and italian honey bee. The thing all these species have in common is that they are all bee species. This is of interest because Neonics seem to be affecting bee species in particular so thats where the research is being focused.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

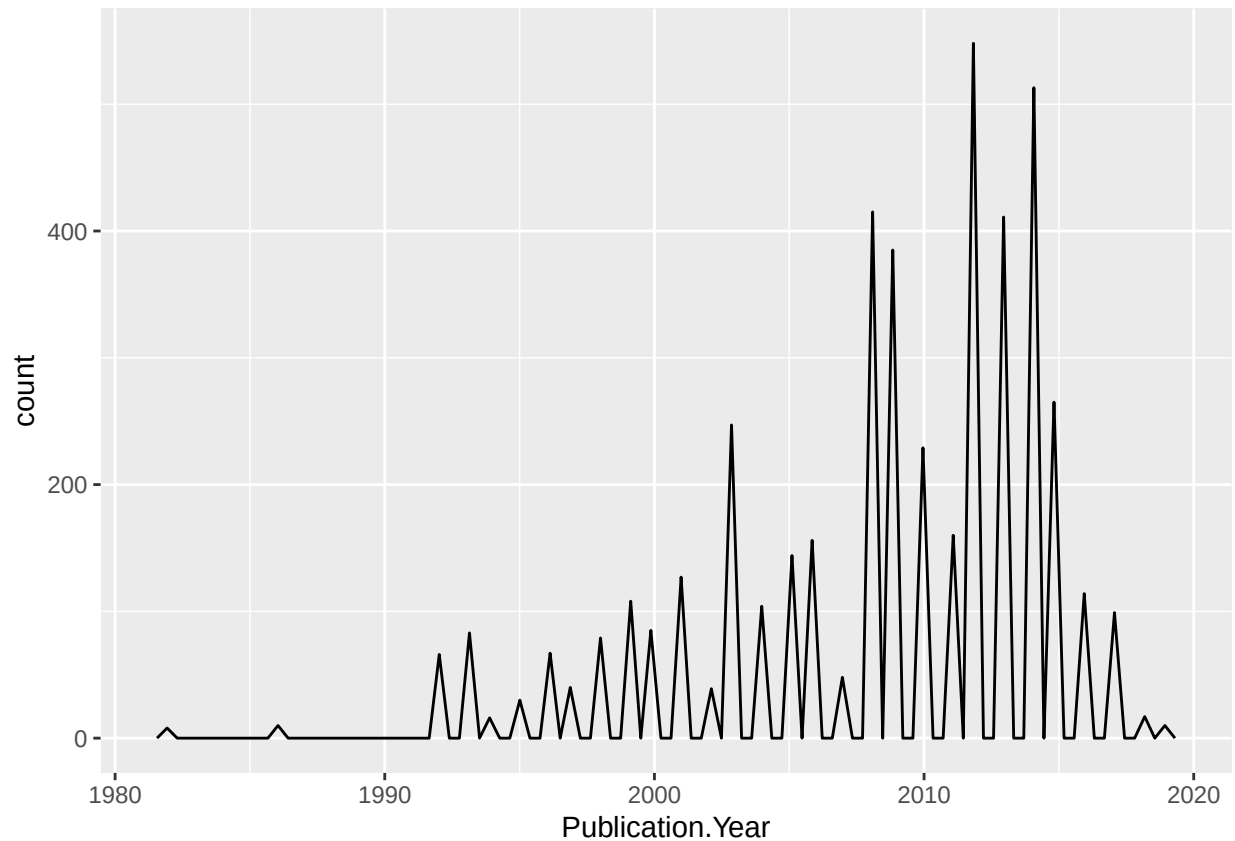
```
## [1] "factor"
```

Answer: Factor, its not numeric because there are some samples with “NR”. meaning not all responses are numeric.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

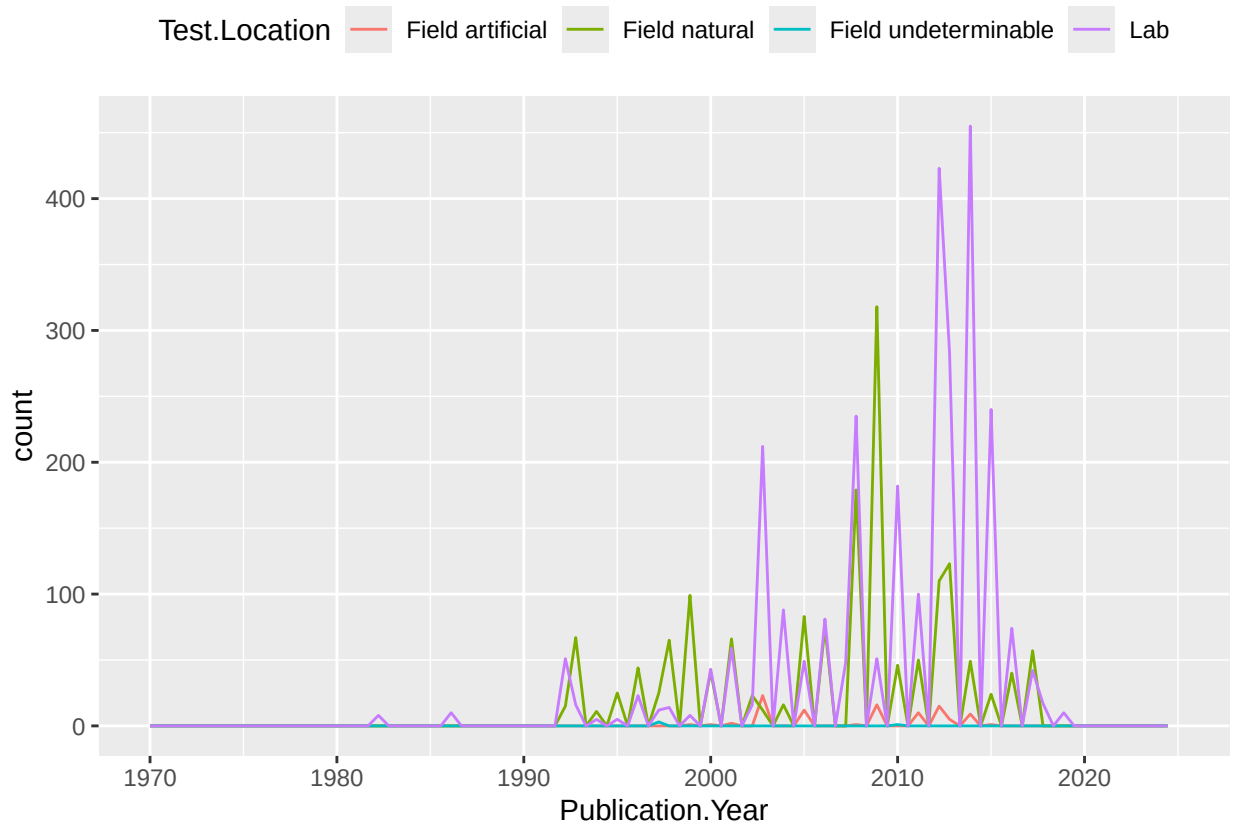
```
ggplot(Neonics) +  
  geom_freqpoly(aes(x = Publication.Year), bins = 100)
```

10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics) +
  geom_freqpoly(aes(x = Publication.Year, color = Test.Location), bins = 100) +
  scale_x_continuous(limits = c(1970, 2025)) +
  theme(legend.position = "top")
```

```
## Warning: Removed 12 rows containing missing values or values outside the scale range
## ('geom_path()').
```



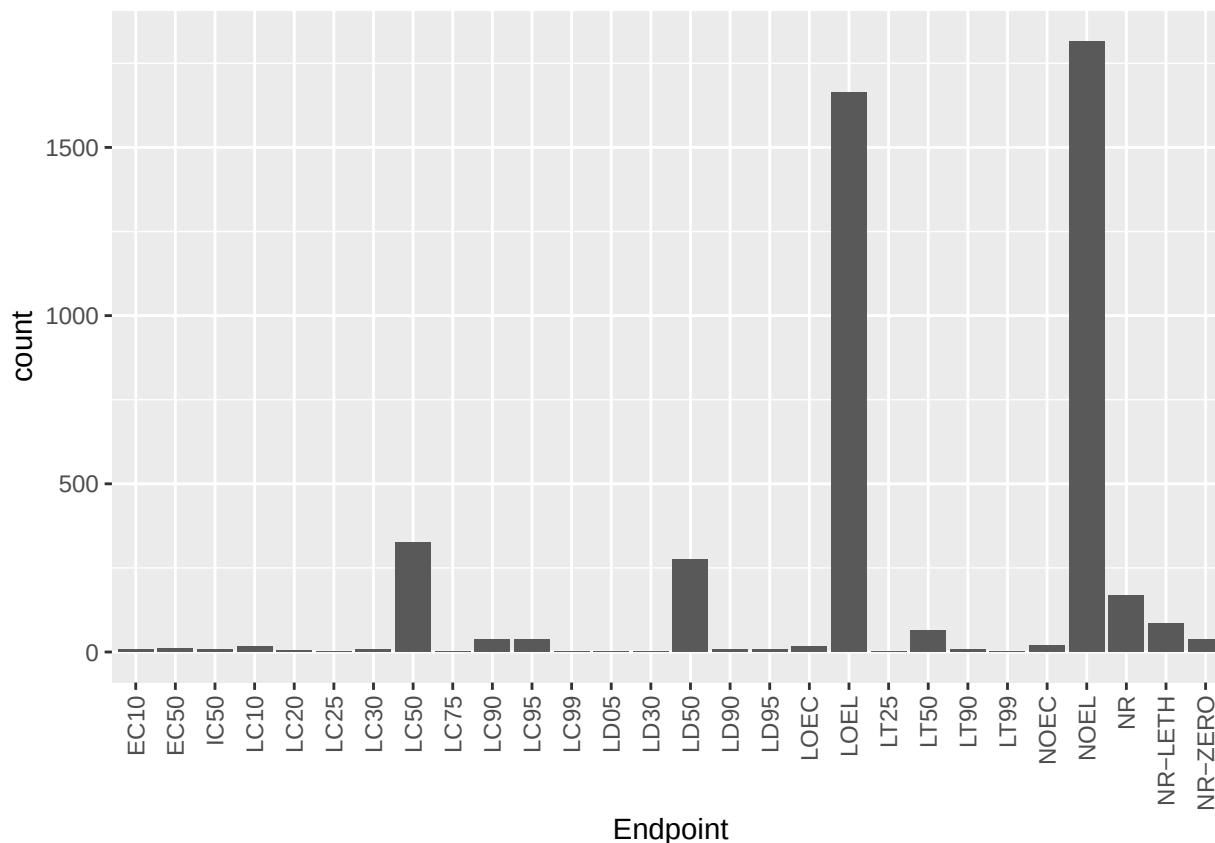
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are Lab and Field natural, the field natural was the most common early on in the studies but lab testing took over as time went on.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(data = Neonics, aes(x = Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common endpoints are NOEL and LOEL. NOEL is defined as no observed effect level and LOEL is defined as lowest observed effect level.

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
unique(Litter$collectDate)
```

```
## [1] 2018-08-02 2018-08-30
## Levels: 2018-08-02 2018-08-30
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

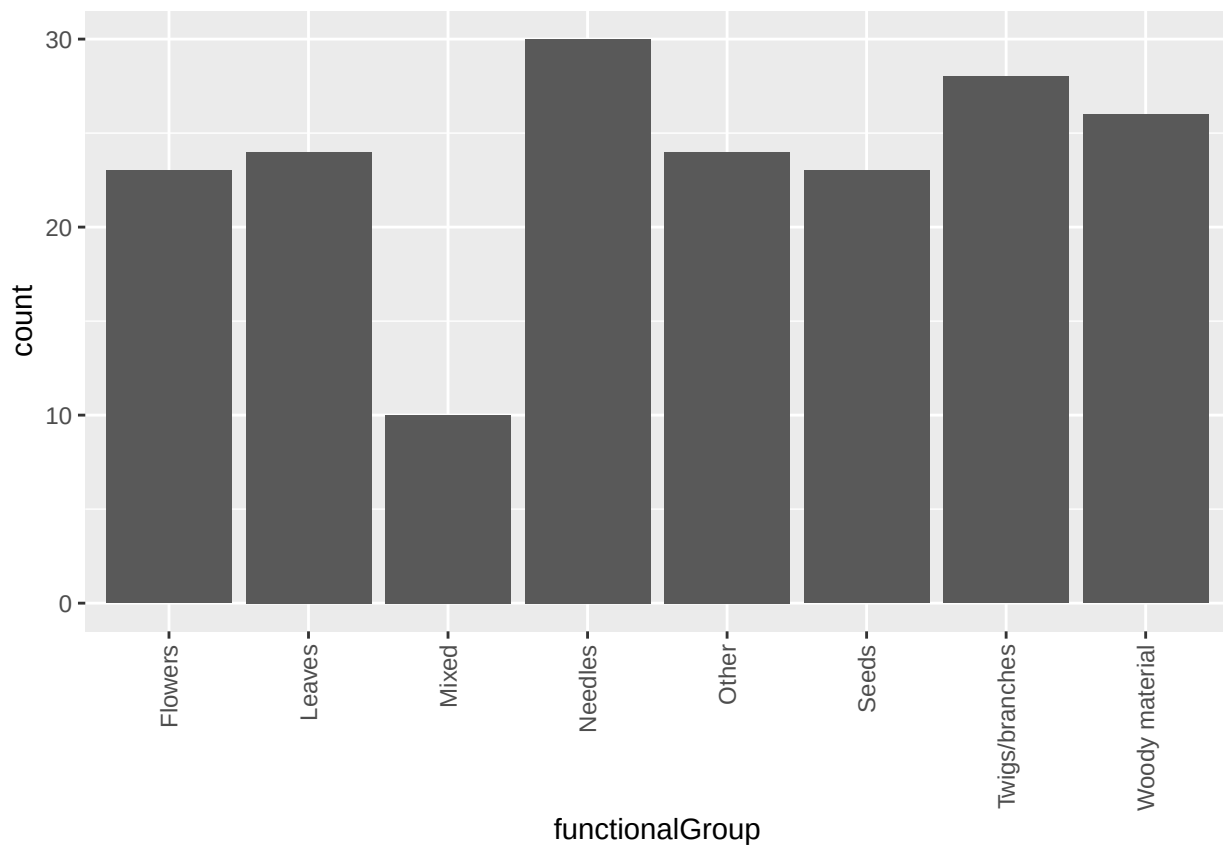
```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: The unique function just lists the different answer types, while sum lists the types and adds up every occurrence of that type.

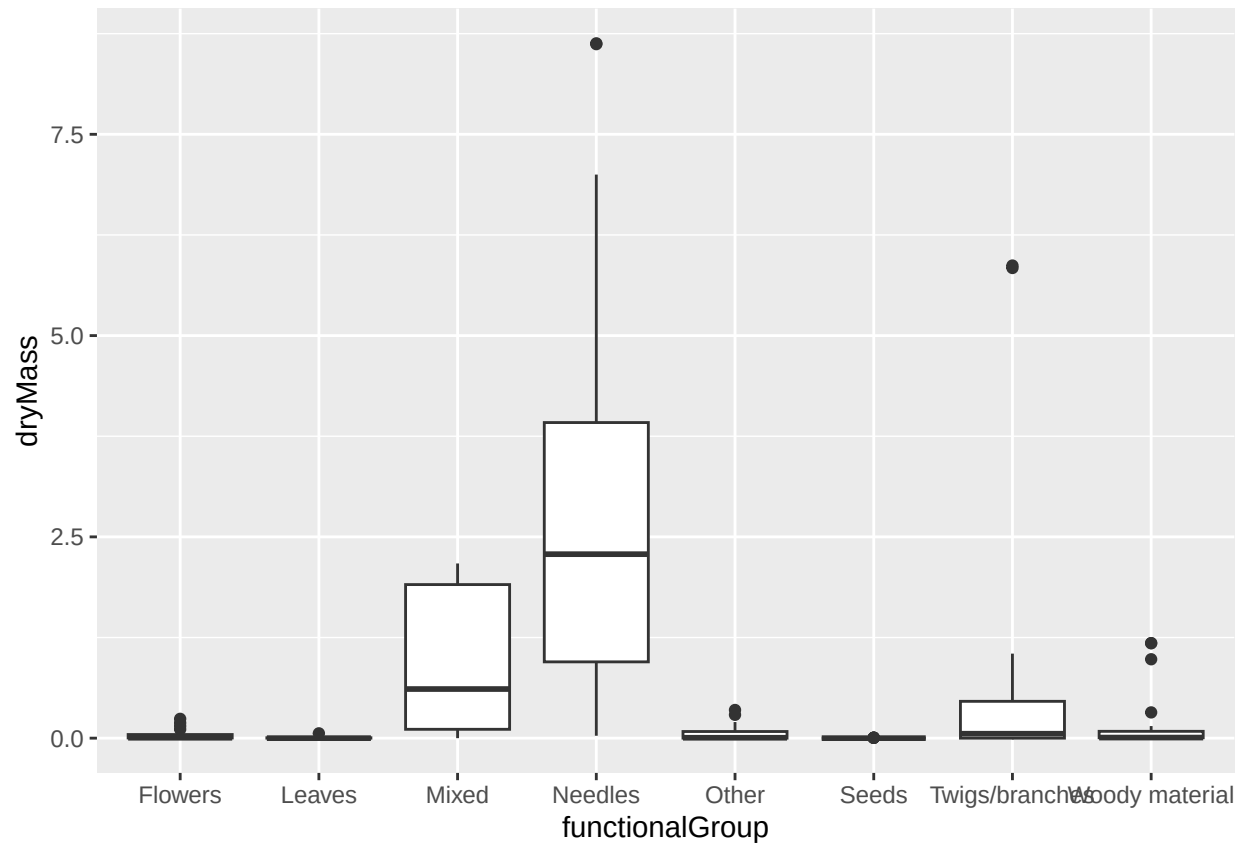
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(data = Litter, aes(x = functionalGroup)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```

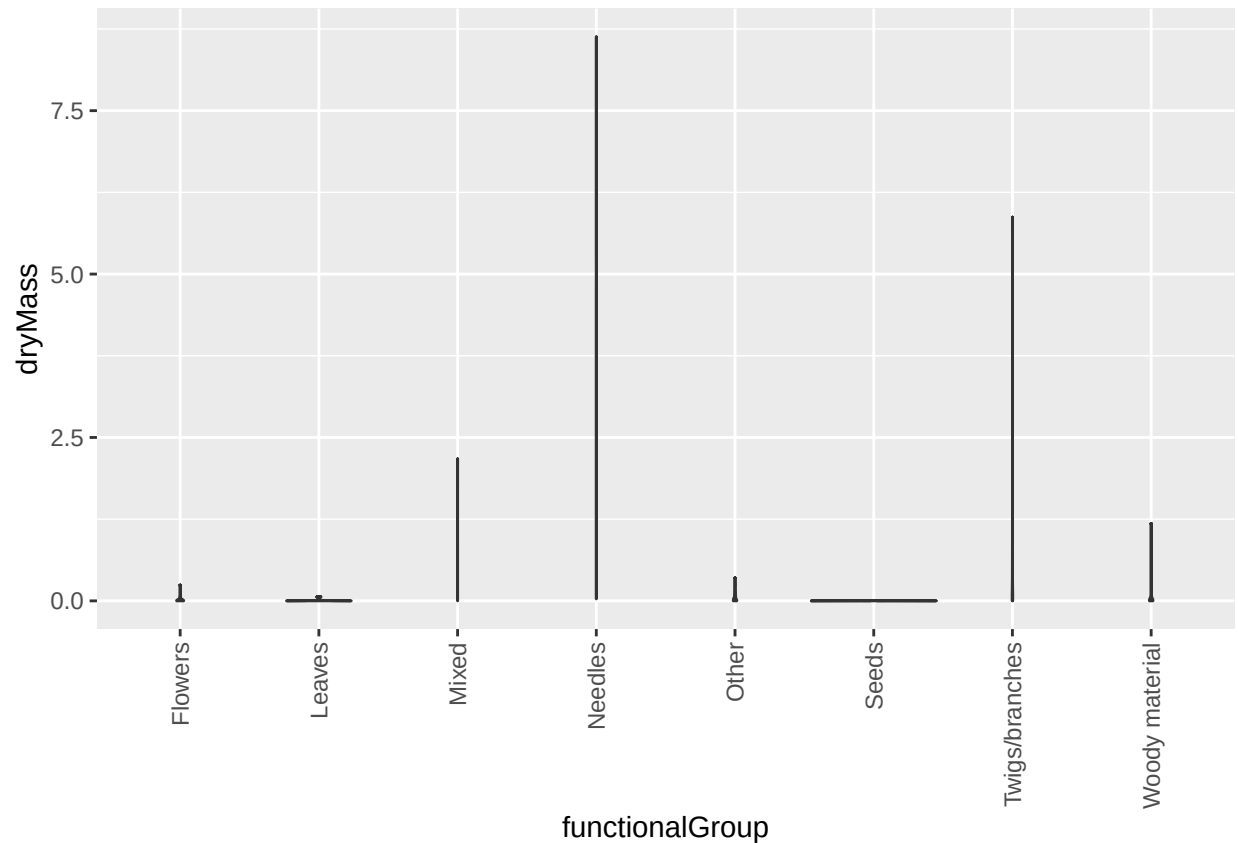


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter) +
  geom_boxplot(aes(x = functionalGroup, y = dryMass))
```



```
ggplot(Litter) +
  geom_violin(aes(x = functionalGroup, y = dryMass),
    draw_quantiles = c(1, 1, 1)) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The box plot is more effective because we can zoom in and actually see the ranges of the function. For the violin plot we are too zoomed out to see the finer details of the function.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles and twigs/branches have the highest biomass at these sites. This indicates that this site is dominated by coniferous trees.